

Puzzles to Puzzle You (Remastered)

Welcome to the eleventh iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Covering Stains) and pose the next one.

As mentioned in the last newsletter, this puzzle comes to us from my dear friend Wei, who first encountered it long ago on a problem set for a statistics course. Today, Wei is a statistics whiz, and he has this puzzle to thank. Until he encountered it, Wei found most of statistics to be a rather dry exercise of mathematical abbreviation—a way of compressing complicated information into tidy summaries. But this puzzle showed him something deeper: Statistics can be a tool for proving existence—a task as philosophically weighty as they come.

In mathematics, this is captured by what is known as the *Probabilistic Method*, pioneered by the inimitable Paul Erdős. The method is uniquely non-constructive. To prove that a particular kind of mathematical object exists, one shows instead that if you pick an object at random from some well-chosen class, the probability that it has the desired property is strictly greater than zero. No explicit example is needed: the positive probability alone guarantees existence with complete certainty. How neat! If only my Erdős Number would go down...



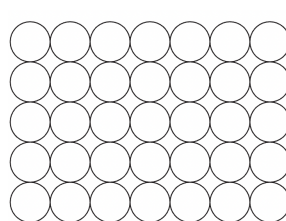
Queen Elizabeth II, accompanied by Sir Hugh Roberts, the Director of the Royal Collection

Puzzle #10 Covering Stains

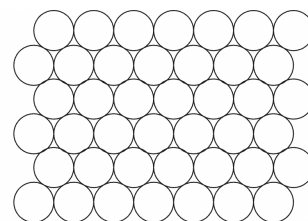
Just as a big event is about to begin, the queen's caterer notices, to his horror, that there are 10 tiny gravy stains on the tablecloth. All he has the time to do is to cover the stains with non-overlapping plates. He has plenty of plates, each a unit disk.

1. Can he do it, no matter how the stains are distributed? Yes
2. How would you try to cover the stains if you had left your glasses at home, and couldn't see where the stains were?

The hexagonal (i.e. densest) packing



square packing



hexagonal packing

The Densest Packing of Circles in $\mathbb{R}^2$ is the Hexagonal Packing

Shout Out (Aaron) Many of you know Aaron as my longtime collaborator and go-to librarian. Lesser-known is the level of Aaron's puzzle grind. He has done nearly every puzzle from this newsletter, but his solution to this one truly astonished me. Not only did Aaron prove that there was a way to cover all ten stains without admitting overlap, but he devised a method for showing that arbitrary numbers of stains are uncoverable. Aaron's method exploits a geometric constraint imposed on circles by their kissing number. From this, he was able to construct an arrangement of 42 stains which are uncoverable. Aaron, unlike Erdős, is unsatisfied with existence. He, like Kant, believes in the thing itself.

Solution It helps to answer the second part of the question before the first. In the case where we do not know the location of the stains, our best bet is to cover as much of the table with

non-overlapping plates as possible. For this task, we need look no further than the densest packing of circles in the plane—the hexagonal packing. Fix some initial plate and carry out the hexagonal packing. Call this t_0 . It can be calculated that t_0 covers 90.7% of the plane.¹

Now, t_0 may work for the second problem, but not the first. Imagine if there are ten stains located in the tiny triangular gaps between circles, then t_0 does not cover them. For this problem, we need to show that in such a case, we can always translate our hexagonal packing in such a way that covers all ten stains.

To prove this, consider the space of all possible translations of the hexagonal packing T . There are infinitely many $t_i \in T$ generated by arbitrary translations. Suppose we chose one of these at random. What is the expected number of stains that t_i will cover?

The answer to this is just 9.07 given that the circles cover 90.7% of the plane. Notice that $9.07 > 9$. But for any $t_i \in T$, t_i can only cover a discrete number of stains from 0 to 10. This implies that there exists some $t' \in T$ such that t' covers all 10 stains. Thus, there exists a translation of the standard hexagonal packing that covers all 10 stains and we are done. $\square$

Why does the last part follow? Why does it follow that if a random translation of the hexagonal tiling is expected to cover 9.07 stains then there exists some translation that covers all ten stains?

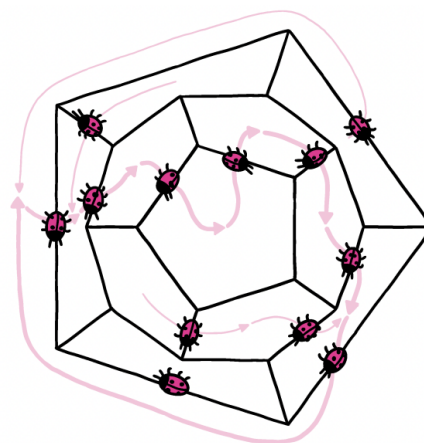
To answer this question, I present an analogy: Suppose there is an exam of 20 questions, each 5 points. Suppose there is no partial credit, so for each question, you either get the full 5 points or 0. If the average exam score was 97.5%, then it's clear that at least one person scored 100%. Otherwise, it wouldn't be possible to get an average of 97.5%.

The same thing happens in our stain problem. We have information about the average or expected value of some distribution, and we can leverage this information to assume something about the distribution itself—namely, that there exists some translation that covers all 10 stains.

Puzzle #11 Bugs on a Polyhedron

Associated with each face of a solid convex polyhedron is a bug which crawls along the perimeter of the face, at varying speed, but only in the clockwise direction.

1. Prove that no schedule will permit all the bugs to circumnavigate their faces and return to their initial positions without incurring a collision.



Bugs on a Regular Dodecahedron

This puzzle is entirely new to me, but I chose it because it reminds me of Edward Frenkel's whimsical discussion of braid groups in his book *Love and Math: The Heart of Hidden Reality* (horrible title, but a serious and brilliant book). Defining a braid group requires some topology, but the idea is simple. Imagine n bugs taking part in a 500 yard dash. Because the bugs are dumb, they don't run straight to the finish. Instead they wander left and right, so that a bug that started to the left of its neighbor might end up on the right. If you follow each bug's path as a 'strand,' the result is a knot-like, interwoven pattern. The set of all such possible patterns forms the braid group.

No other image from a math book has stuck in my mind quite like that one. While our bug puzzle doesn't involve braid groups, it still invites us to imagine the complex, random journeys of tiny, dumb creatures. A delightfully Sisyphean thought.

¹The ratio of the area of a circle to its circumscribed hexagon is $\frac{\pi\sqrt{3}}{6}$ to be exact.